

ASSIGNMENT REPORT-5

Part 1 Draft: Maximum Likelihood Parameter Estimation

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For the subject

Pattern Recognition and Machine Learning

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PART 1: SYNTHETIC GAUSSIAN DATA

[WRITE HERE: Introduce the goal of estimating Gaussian parameters and classifying held-out points in your own words.]

Experimental setup

Three classes were generated with means $\boldsymbol{\mu}_1 = (0, 0)$, $\boldsymbol{\mu}_2 = (4, 0)$, and $\boldsymbol{\mu}_3 = (2, 4)$. Each class contains 10,000 two-dimensional points. The split is 8,000 training and 2,000 test points per class. The Gaussian generator uses the Marsaglia polar method from the earlier labs. All six cases use equal class counts.

Parameter estimation and classification

For class k , the maximum likelihood mean estimate is

$$\hat{\boldsymbol{\mu}}_k = \frac{1}{N_k} \sum_{i=1}^{N_k} \mathbf{x}_{ki}.$$

The full class covariance estimate uses the training residuals:

$$\hat{\boldsymbol{\Sigma}}_k = \frac{1}{N_k} \sum_{i=1}^{N_k} (\mathbf{x}_{ki} - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_{ki} - \hat{\boldsymbol{\mu}}_k)^\top.$$

For a shared covariance, the residual outer products are pooled across classes and divided by the total number of training points. A diagonal estimate keeps only the two diagonal entries. An isotropic estimate uses half the trace as its common variance. The denominator is N_k or the total training count, as appropriate for maximum likelihood estimation.

The test point is assigned to the class with the largest Gaussian log score:

$$g_k(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \hat{\boldsymbol{\mu}}_k)^\top \hat{\boldsymbol{\Sigma}}_k^{-1}(\mathbf{x} - \hat{\boldsymbol{\mu}}_k) - \frac{1}{2} \log |\hat{\boldsymbol{\Sigma}}_k|.$$

The prior is equal for all three classes (1/3), so its log is the same constant in every score. Confusion matrix rows below are true classes; columns are predicted classes.

Accuracy summary

Table 1: Test accuracy for the six synthetic cases (6,000 test points per case).

Covariance case	Accuracy (%)
Shared isotropic	97.20
Different isotropic	89.05
Shared diagonal	82.73
Different diagonal	87.37
Shared full	94.45
Different full	87.53

Shared isotropic

Table 2: Generating values and training estimates for shared isotropic.

Class	Estimated mean	Generating covariance	Estimated covariance
1	$(-0.007, 0.016)$	$\begin{bmatrix} 1.000 & 0.000 \\ 0.000 & 1.000 \end{bmatrix}$	$\begin{bmatrix} 1.005 & 0.000 \\ 0.000 & 1.005 \end{bmatrix}$
2	$(4.010, -0.002)$	$\begin{bmatrix} 1.000 & 0.000 \\ 0.000 & 1.000 \end{bmatrix}$	$\begin{bmatrix} 1.005 & 0.000 \\ 0.000 & 1.005 \end{bmatrix}$
3	$(2.001, 4.003)$	$\begin{bmatrix} 1.000 & 0.000 \\ 0.000 & 1.000 \end{bmatrix}$	$\begin{bmatrix} 1.005 & 0.000 \\ 0.000 & 1.005 \end{bmatrix}$

Table 3: Confusion matrix for shared isotropic.

True \ Predicted	1	2	3
1	1934	45	21
2	35	1944	21
3	18	28	1954

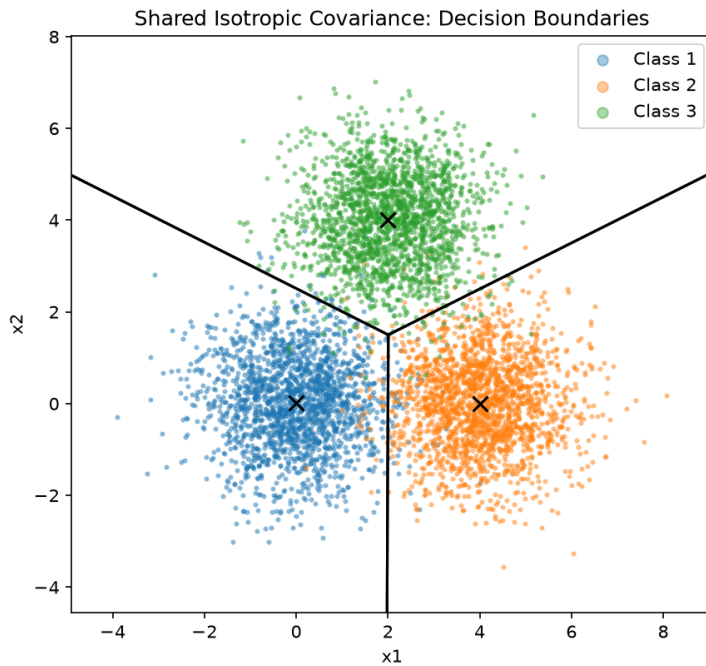


Figure 1: Decision boundaries with test points for shared isotropic.

Test accuracy: **97.20%**.

Different isotropic

Table 4: Generating values and training estimates for different isotropic.

Class	Estimated mean	Generating covariance	Estimated covariance
1	(0.016, 0.006)	$\begin{bmatrix} 1.000 & 0.000 \\ 0.000 & 1.000 \end{bmatrix}$	$\begin{bmatrix} 0.984 & 0.000 \\ 0.000 & 0.984 \end{bmatrix}$
2	(4.020, -0.006)	$\begin{bmatrix} 2.000 & 0.000 \\ 0.000 & 2.000 \end{bmatrix}$	$\begin{bmatrix} 2.042 & 0.000 \\ 0.000 & 2.042 \end{bmatrix}$
3	(2.051, 4.008)	$\begin{bmatrix} 4.000 & 0.000 \\ 0.000 & 4.000 \end{bmatrix}$	$\begin{bmatrix} 4.068 & 0.000 \\ 0.000 & 4.068 \end{bmatrix}$

Table 5: Confusion matrix for different isotropic.

True \ Predicted	1	2	3
1	1872	75	53
2	106	1776	118
3	118	187	1695

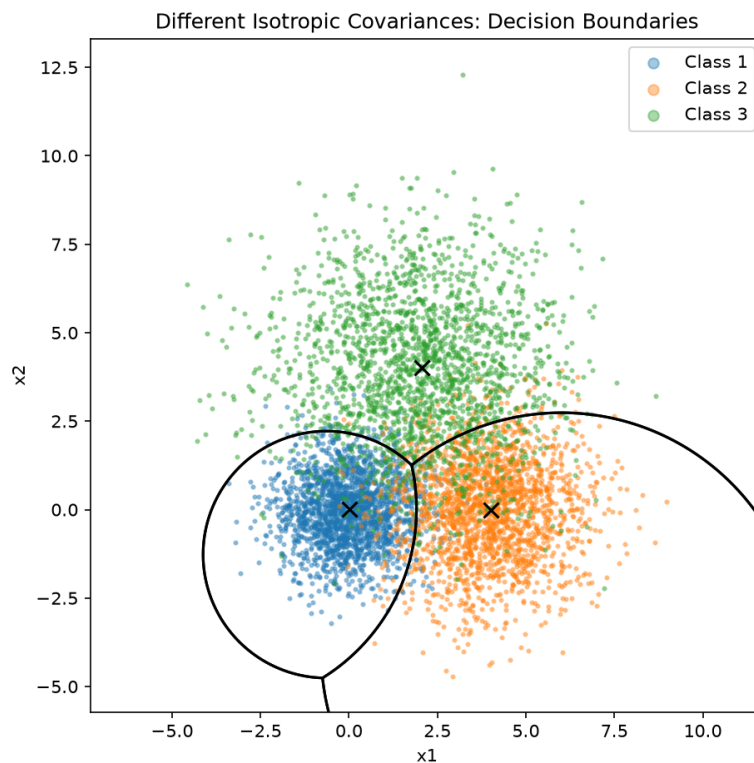


Figure 2: Decision boundaries with test points for different isotropic.

Test accuracy: **89.05%**.

Shared diagonal

Table 6: Generating values and training estimates for shared diagonal.

Class	Estimated mean	Generating covariance	Estimated covariance
1	(0.007, -0.011)	$\begin{bmatrix} 2.000 & 0.000 \\ 0.000 & 4.000 \end{bmatrix}$	$\begin{bmatrix} 1.999 & 0.000 \\ 0.000 & 3.965 \end{bmatrix}$
2	(4.011, 0.047)	$\begin{bmatrix} 2.000 & 0.000 \\ 0.000 & 4.000 \end{bmatrix}$	$\begin{bmatrix} 1.999 & 0.000 \\ 0.000 & 3.965 \end{bmatrix}$
3	(1.997, 3.965)	$\begin{bmatrix} 2.000 & 0.000 \\ 0.000 & 4.000 \end{bmatrix}$	$\begin{bmatrix} 1.999 & 0.000 \\ 0.000 & 3.965 \end{bmatrix}$

Table 7: Confusion matrix for shared diagonal.

True \ Predicted	1	2	3
1	1671	135	194
2	109	1712	179
3	218	201	1581

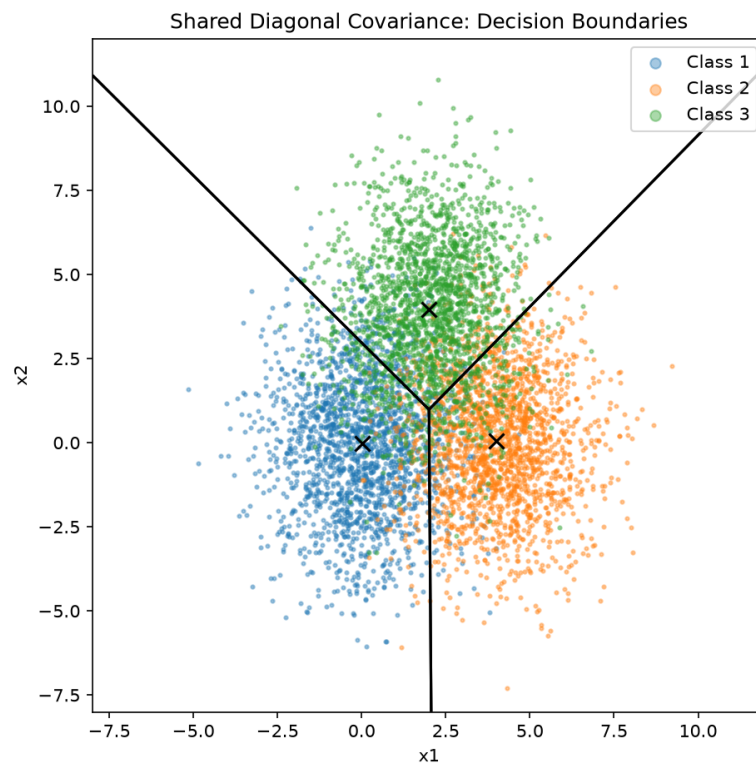


Figure 3: Decision boundaries with test points for shared diagonal.

Test accuracy: **82.73%**.

Different diagonal

Table 8: Generating values and training estimates for different diagonal.

Class	Estimated mean	Generating covariance	Estimated covariance
1	(0.017, 0.006)	$\begin{bmatrix} 1.000 & 0.000 \\ 0.000 & 4.000 \end{bmatrix}$	$\begin{bmatrix} 1.003 & 0.000 \\ 0.000 & 4.095 \end{bmatrix}$
2	(3.991, -0.007)	$\begin{bmatrix} 4.000 & 0.000 \\ 0.000 & 1.000 \end{bmatrix}$	$\begin{bmatrix} 4.066 & 0.000 \\ 0.000 & 1.006 \end{bmatrix}$
3	(1.999, 4.024)	$\begin{bmatrix} 2.000 & 0.000 \\ 0.000 & 3.000 \end{bmatrix}$	$\begin{bmatrix} 1.996 & 0.000 \\ 0.000 & 3.027 \end{bmatrix}$

Table 9: Confusion matrix for different diagonal.

True \ Predicted	1	2	3
1	1762	106	132
2	192	1746	62
3	161	105	1734

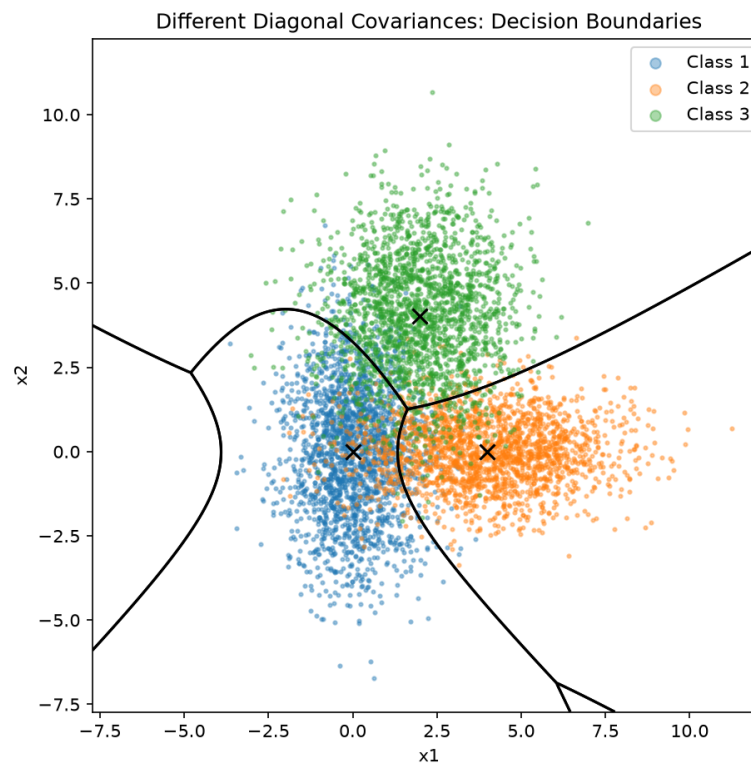


Figure 4: Decision boundaries with test points for different diagonal.

Test accuracy: **87.37%**.

Shared full

Table 10: Generating values and training estimates for shared full.

Class	Estimated mean	Generating covariance	Estimated covariance
1	$(-0.014, 0.008)$	$\begin{bmatrix} 2.000 & -1.500 \\ -1.500 & 2.000 \end{bmatrix}$	$\begin{bmatrix} 2.004 & -1.505 \\ -1.505 & 2.009 \end{bmatrix}$
2	$(4.062, -0.042)$	$\begin{bmatrix} 2.000 & -1.500 \\ -1.500 & 2.000 \end{bmatrix}$	$\begin{bmatrix} 2.004 & -1.505 \\ -1.505 & 2.009 \end{bmatrix}$
3	$(2.029, 3.990)$	$\begin{bmatrix} 2.000 & -1.500 \\ -1.500 & 2.000 \end{bmatrix}$	$\begin{bmatrix} 2.004 & -1.505 \\ -1.505 & 2.009 \end{bmatrix}$

Table 11: Confusion matrix for shared full.

True \ Predicted	1	2	3
1	1962	38	0
2	40	1828	132
3	2	121	1877

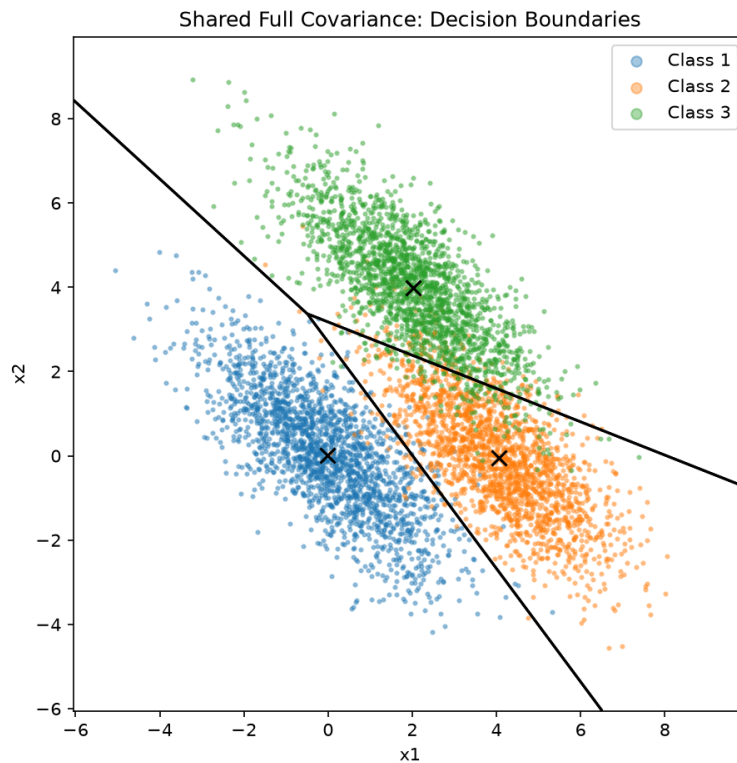


Figure 5: Decision boundaries with test points for shared full.

Test accuracy: **94.45%**.

Different full

Table 12: Generating values and training estimates for different full.

Class	Estimated mean	Generating covariance	Estimated covariance
1	(0.021, 0.003)	$\begin{bmatrix} 2.000 & 1.300 \\ 1.300 & 1.500 \end{bmatrix}$	$\begin{bmatrix} 2.004 & 1.293 \\ 1.293 & 1.477 \end{bmatrix}$
2	(3.995, -0.002)	$\begin{bmatrix} 3.000 & -1.500 \\ -1.500 & 1.500 \end{bmatrix}$	$\begin{bmatrix} 3.002 & -1.510 \\ -1.510 & 1.512 \end{bmatrix}$
3	(2.018, 3.983)	$\begin{bmatrix} 1.200 & 0.700 \\ 0.700 & 3.000 \end{bmatrix}$	$\begin{bmatrix} 1.173 & 0.664 \\ 0.664 & 2.953 \end{bmatrix}$

Table 13: Confusion matrix for different full.

True \ Predicted	1	2	3
1	1756	108	136
2	175	1709	116
3	153	60	1787

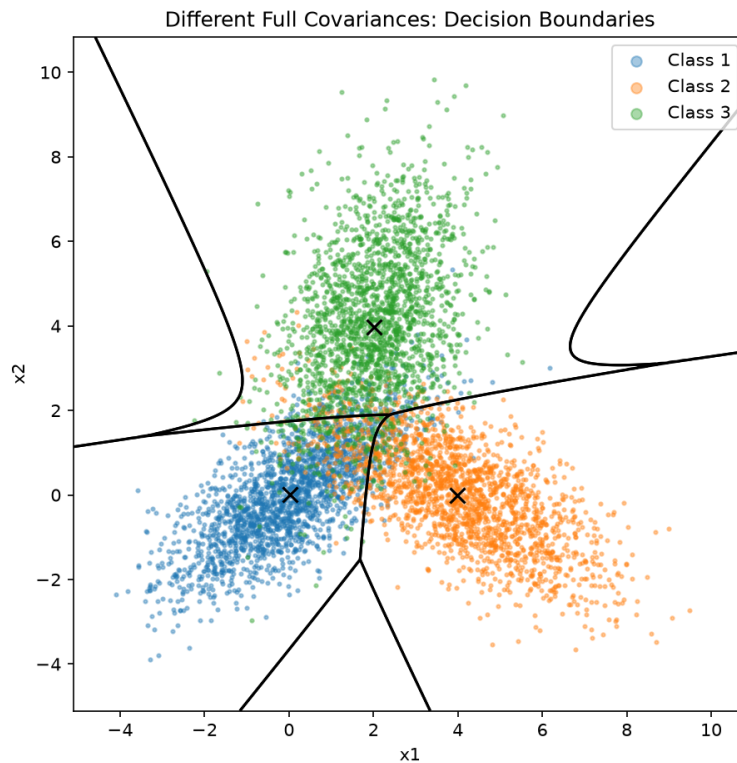


Figure 6: Decision boundaries with test points for different full.

Test accuracy: **87.53%**.

OBSERVATIONS

[WRITE HERE: *Describe how each covariance type changes the point clouds and decision boundaries. Compare where the confusion matrices show overlap. Explain what the estimated parameters tell you.*]